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CALCIUM HOMEOSTASIS

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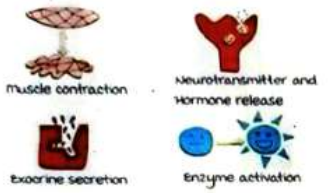
Normal calcium levels : 8.7–10.2 mg/dL.

Distribution & Absorption of Ca^{2+} 00:00:44

DISTRIBUTION OF TOTAL CALCIUM :

Calcium (Ca^{2+})

Free form (50%)
- Ionic calcium.
- Most active.
- Responsible for actions.



Free calcium actions

Bound form (41%)
- Bound to albumin.
- Storage form.
- Not active.

Complex form (9%)
- Citrate.
- Phosphate.

Applied aspect :

Hypocalcemic tetany :

- Concept : Ratio of free to bound calcium determined by :
  - Blood pH.
  - Albumin (Binds both H^+ & Calcium).
- Mechanism :

↓ Ca^{2+} → Na^+ channels open more frequently → ↑ depolarisation → ↑ frequency of action potentials → Hyperexcitability → Hypocalcemic tetany.

- Signs :
  1. Trousseau's sign : Eliciting carpopedal spasm by ↑20 mm Hg above SBP for 3 min.
  2. Chovtsek's sign : Facial muscle twitching on tapping facial nerve at angle of jaw.

} D/t hyperexcitability



Trousseau's sign

- Eg : Respiratory alkalosis → ↓ H^+ levels → ↑ binding of Ca^{2+} to albumin → ↓ free Ca^{2+}.

ABSORPTION OF CA^{2+}

Site : Duodenum & proximal jejenum.

Factors affecting :

↑ Calcium absorption	↓ Calcium absorption
• Gastric acid • Protein-rich diet	• Phytates • Oxalates • Fatty acids } Form insoluble Ca^{2+} complexes

What determines the ratio of free and bound calcium ?

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Hormones Regulating Ca^{2+} Homeostasis

Three hormone three organ model :

Three hormones :

- Parathormone
- Vitamin D
- Calcitonin ($\downarrow Ca^{2+}$)

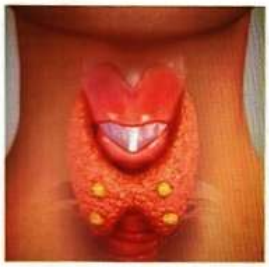
($\uparrow Ca^{2+}$)

Three organs :

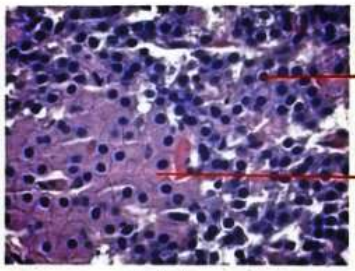
- Bone (Bank for Ca^{2+})
- GIT (Absorption of dietary Ca^{2+})
- Kidneys

PARATHORMONE (PTH) :

Site :



Parathyroid gland

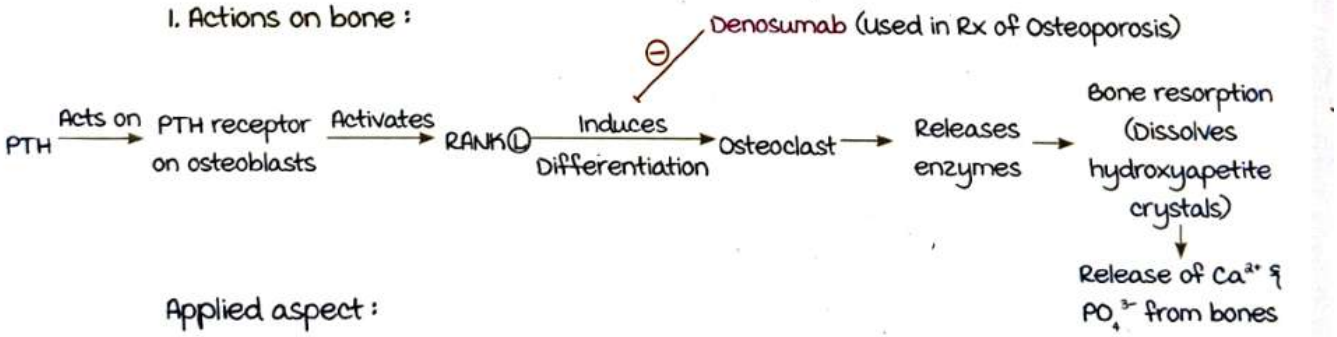


Histology

Structure : 84 amino acids (Large protein).

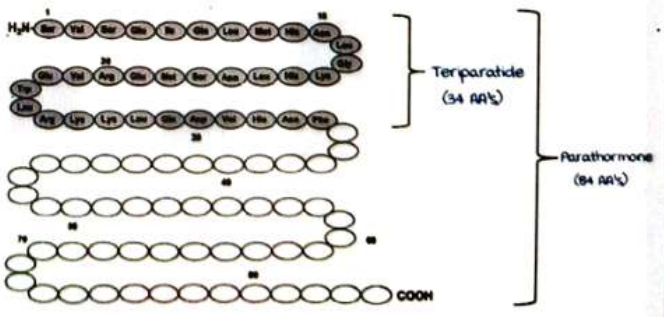
Actions :

1. Actions on bone :



Applied aspect :

- Concept :
  - Osteoblast : Bone formation.
  - Osteoclast : Bone resorption.



Teriparatide

- Use :

  Teriparatide : Drug form of PTH.
  - Continuous administration → Bone resorption.
  - Intermittent administration → Bone formation.

Feedback

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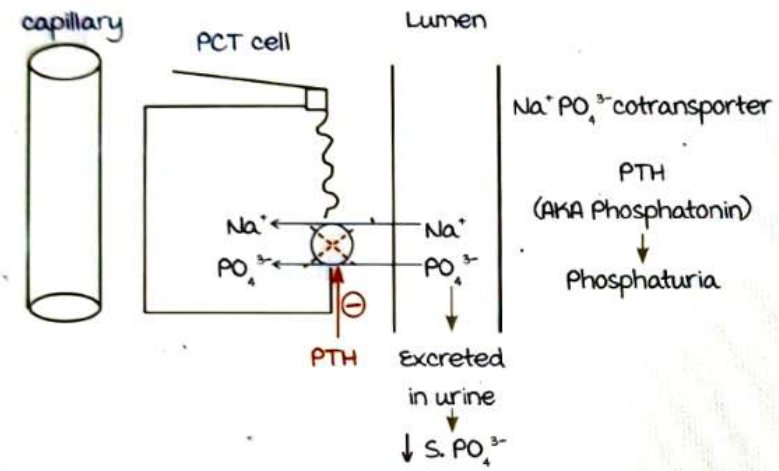
2. Action on Kidneys :

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PCT

DCT

↑ Ca^{2+} reabsorption



Overall action of PTH :
- ↑ S. Ca^{2+}
- ↓ S. PO_4^{3-}

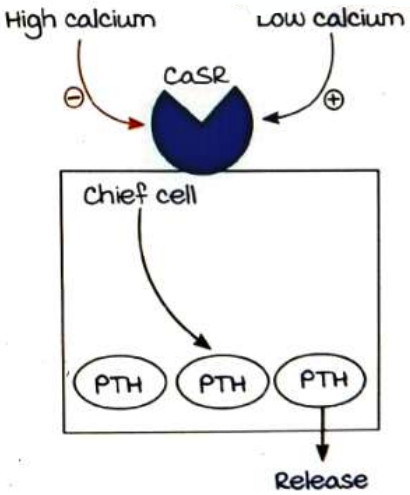
3. Action on GIT :

PTH ⟶ ↑ Vit.D ⟶ ↑↑ Ca^{2+} absorption from GIT (Indirect action).

Calcium sensing receptors (CaSR) :

Applied aspect

Mutation of CaSR ⟶ PTH not suppressed ⟶ PTH ↑↑↑ ⟶ Familial hypocalciuric hypercalcemia



VITAMIN D :

Synthesis :

7- Dehydrocholesterol (Skin)

↓ UV-B rays (280-320nm)

Vit D3 or Cholecalciferol

↓ 25-hydroxylase in liver

25-hydroxycholecalciferol/calcidiol (Lab investigation for Vit.D status)

↓ Rate limiting step : 1α- Hydroxylase in Kidney

1,25-Dihydroxycholecalciferol/calcitriol (most active form)

Actions :

1. On GIT	2. On Bone	3. On Kidneys
Ca^{2+} absorption from duodenum ↓ • ↑ Calbindin. • ↑ expression of Ca^{2+} ATPase. • ↑ Alkaline phosphatase.	Dual action • Low calcium states : Bone resorption (Vitamin D receptors on osteoblast ⟶ Action similar to PTH). • High calcium states : Bone formation.	↑resorption of Ca^{2+} & PO_4^{3-} ↓ ↑ S.Ca^{2+} & ↑ S. PO_4^{3-}

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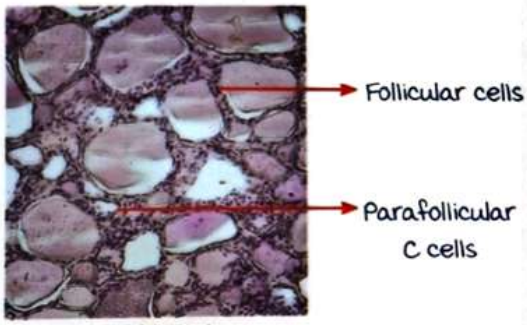
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CALCITONIN :

Source :

Parafollicular c cells of thyroid gland.

Note : Calcitonin is a tumor marker for medullary carcinoma of thyroid (Originates from parafollicular c cells).



Thyroid histology

Actions :

Calcium lowering hormone :

- Bone : Calcitonin receptor on osteoclast → Inhibits osteoclast activity → ↓ bone resoprtion
- GIT : ↓ absorption of Ca^{2+}
- Kidney : ↓ reabsorption of Ca^{2+}

Applied aspect :

1. Role of pregnancy & lactation :
   - Concept : In pregnancy → ↑ Ca^{2+} demand → Ca^{2+} utilized from bone reserve.
   - Regulation : Calcitonin prevents excessive bone loss.
2. Calcitonin analogues :
   - Derived from : Salmon fish.
   - Drug name : Salcatonin.

Note :

Paraneoplastic syndrome : Hypercalcemia of malignancy.

- D/t parathormone related peptide (PTH_{RP}).
- Associated with :
  - Squamous cell carcinoma of lung.
  - Small cell carcinoma of lung.

Bone Markers

00:48:10

Bone formation markers	Bone resorption markers
• Osteocalcin. • Alkaline phosphatase. • Pro-peptides of type I collagen.	• Blood – TRAP (Tartrate resistant acid phosphatase). – mmp (matrix metalloproteinase). – Cathepsin K. • Urine – NTX (N telopeptides of collagen crosslinks). – CTX (C telopeptides of collagen crosslinks).

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Summary

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	PTH	Vitamin D	Calcitonin
Stimulus	↓ in serum Ca^{2+}	• ↓ in serum Ca^{2+}. • ↓ in serum PO_4^{3-}. • ↑ PTH.	↑ in serum Ca^{2+}
Bone	↑ bone resorption	Both bone formation & resorption	Inhibits bone resorption
Intestine	Indirect action: ↑ Vitamin D	↑ Ca^{2+} & ↑ PO_4^{3-} absorption	↓ Ca^{2+} absorption
Kidney	• ↑ Ca^{2+} reabsorption. • ↑ PO_4^{3-} excretion.	↑ Ca^{2+} & ↑ PO_4^{3-} reabsorption	↑ Ca^{2+} excretion
Overall effect	• ↑ S. Ca^{2+}. • ↓ S. PO_4^{3-}.	• ↑ S. Ca^{2+}. • ↑ S. PO_4^{3-}.	• ↓ S. Ca^{2+}.

Feedback